

Clean-Label Poisoning on a Four-Class CIFAR-10 Subset

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Abstract—We extend clean-label poisoning to a four-class CIFAR-10 subset (Plane, Car, Bird, Cat). A small CNN trained on clean data reaches 73.5% test accuracy. After inserting ten feature-collision poisons labeled as Bird, the retrained model maintains 74.3% accuracy yet misclassifies 3/10 targets as Bird. The report summarizes the dataset setup, poison optimization procedure, and the observed trade-off between stealth and attack success.

I. INTRODUCTION

Clean-label poisoning forces misclassification without tampering with ground-truth labels, making it relevant to biometric pipelines that trust curated data. This experiment builds the attack on CIFAR-10 with 500 images per class. A modest perturbation budget ($\epsilon = 0.1$) is enough to redirect some plane targets into the bird class while keeping aggregate accuracy roughly stable.

II. METHODOLOGY

A. Data Preparation

The data preparation keeps 500 samples each from Plane, Car, Bird, and Cat. Ten plane targets are removed from training and saved for later evaluation. Ten bird bases remain in the clean set and are paired with the targets for poisoning. The resulting training split contains 1,990 samples, and the test split uses 100 images per class.

B. Model Architecture and Baseline Training

We use SimpleCNN (three convolutional blocks, two fully connected layers) and train with Adam ($\text{lr} = 0.001$, batch size 32, 15 epochs). The clean model achieves 73.5% accuracy on the held-out test subset.

C. Poison Generation

Feature-collision poisons are produced by minimizing $\|f(x) - f(t)\|_2^2$ between current poison features and target features while clipping each pixel deviation to ± 0.1 around the base bird. Optimization runs for 200 iterations with SGD ($\text{lr} = 0.1$). Snapshots every 20 iterations show that poisons stay visually bird-like while moving toward the plane feature manifold.

D. Retraining with Poisons

We append the ten poisons (labeled Bird) to the clean training set and retrain SimpleCNN from scratch for 15 epochs. Evaluation uses overall accuracy, per-class accuracy, and target predictions to measure collateral damage and attack success.

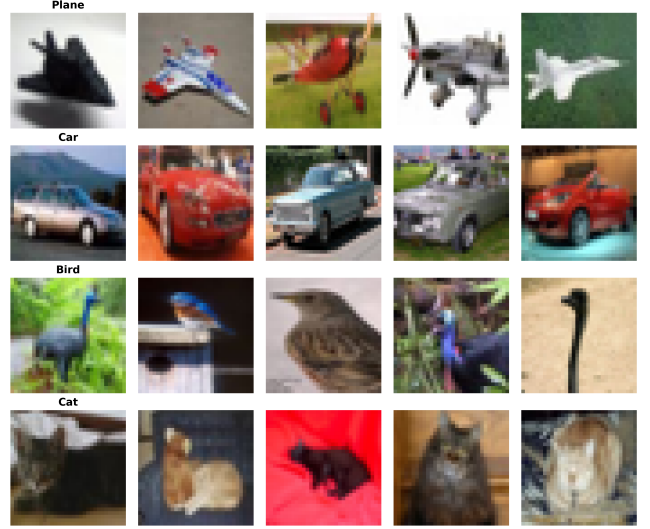


Fig. 1. Sample images from each of the four classes: Plane, Car, Bird, Cat.



Fig. 2. Held-out plane targets (row 1) and bird bases (row 2).

III. RESULTS

A. Baseline Metrics

Figure 1 shows samples from the four selected classes. Figure 2 shows the target/base pairs. Figure 3 shows the clean and poisoned training loss curves. Figure 4 compares confusion matrices before and after poisoning.

B. Poison Characteristics

Figure 5 shows the first poison over optimization and compares final poisons against their base birds. Figure 6 shows the feature distance decreasing during optimization. Figure 7 shows the magnified perturbations added to base images.

C. Attack Impact

Table I summarizes headline metrics. Overall accuracy remains stable, yet three of ten target planes now map to Bird. Figure 8 shows per-target predictions before and after retraining. The successful flips are targets that were already closer to Bird than the others.

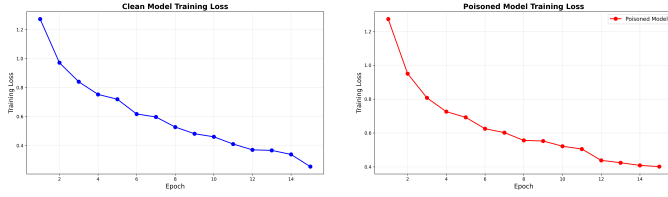


Fig. 3. Training loss curves for clean model (left) and poisoned model (right).

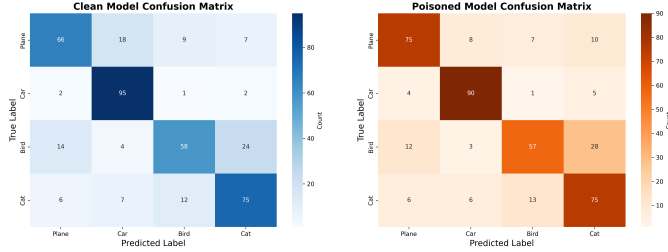


Fig. 4. Confusion matrices before (left) and after (right) poisoning.

IV. DISCUSSION

Only 30% of targets flipped. The perturbation budget was small, the CNN was limited, and the run used only ten poisons. Larger ϵ , more optimization steps, or more poison images could raise success, but the poisons would become easier to detect. Conversely, data sanitization that monitors feature outliers may defeat this attack even when labels remain consistent.

V. CONCLUSION

This clean-label poisoning pipeline keeps global accuracy stable while compromising part of the target set. With a lightweight CNN and a low visual budget, the attack flipped three selected plane images. That is enough to show why data ingestion checks matter when models rely on external training data.

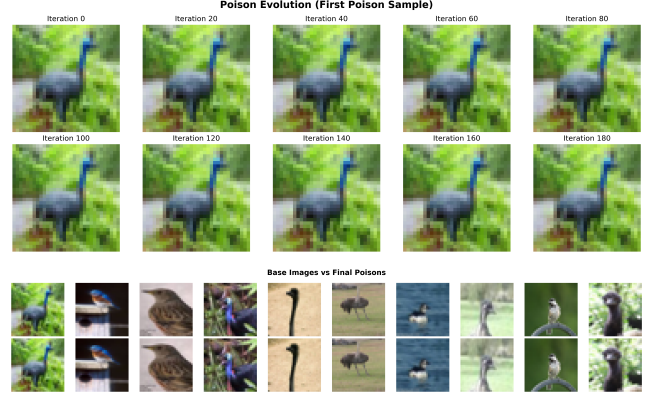


Fig. 5. Poison evolution snapshots (top) and final poisons compared to their base birds (bottom).

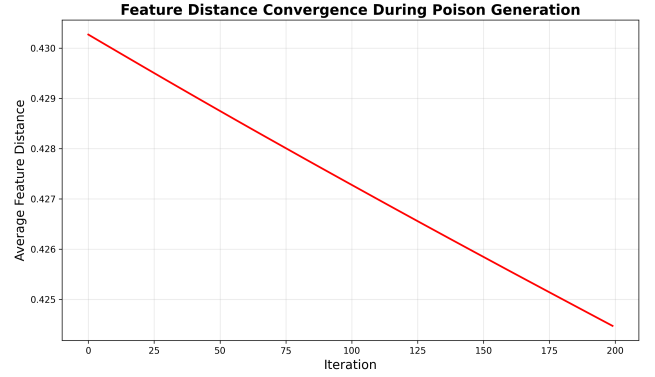


Fig. 6. Feature distance convergence during poison generation.

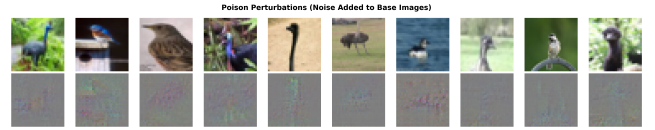


Fig. 7. Poison perturbations (10x magnified). Top: base images. Bottom: added perturbation pattern.

TABLE I
CLEAN VS. POISONED PERFORMANCE.

Metric	Clean	Poisoned
Test accuracy	73.5 %	74.3 %
Plane accuracy	78.0 %	75.0 %
Bird accuracy	60.0 %	57.0 %
Target success	0/10	3/10

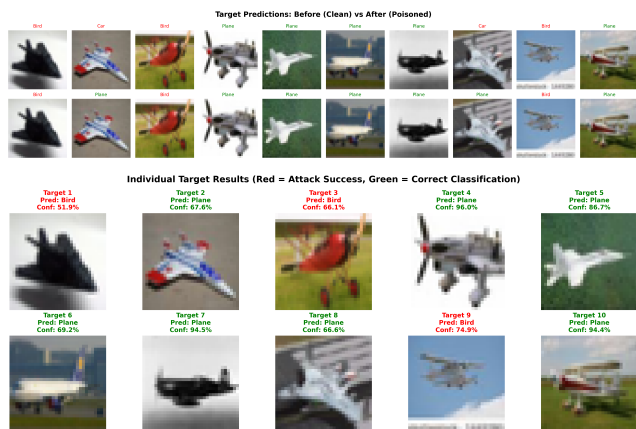


Fig. 8. Top: predictions on each target before (clean) and after poisoning. Bottom: individual target results with confidence scores (red = attack success, green = correct).

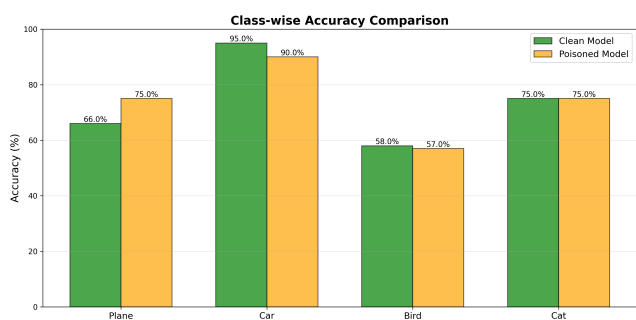


Fig. 9. Per-class accuracy comparison between clean and poisoned models.

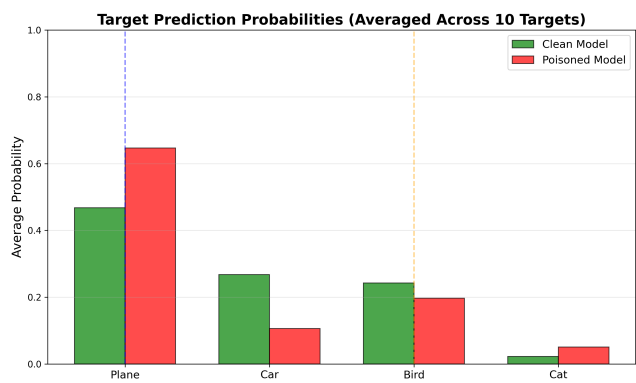


Fig. 10. Average prediction probabilities on targets before and after poisoning.